

Fixed length Subnet mask /16 /18 /19
 Variable length Subnet mask /17 /10 /26 ✖
 Class full 10.0.0.0 /8
 Class B 10.0.0.0 /20 /16

TCP/IP Stack

OSI Model	TCP/IP Stack
Application	Application
Presentation	
Session	
Transport	Transport
Network	Internet
Data Link	
Physical	Network Access

Port No

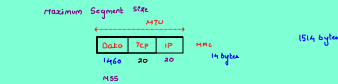
Source identification

Protocol No

Protocol Number	Protocol
1	TCP
6	TCP
17	TCP
18	TCP
20	TCP
21	TCP

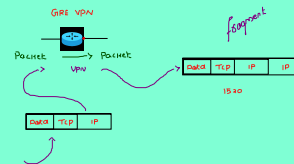
MTU = 1500 bytes

MSS



Segment / Fragment

MTU = 1500



Open → Three-way handshake

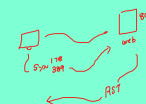
Close → Four-way handshake

Tcp header

Source port	Destination Port
Sequence number	
Acknowledgment number	
Window	
Checksum	
Urgent pointer	

(Error detection)

SYN flag
 ACK flag
 RST flag
 FIN flag



UDP header

Source port	Destination port
Length	
Checksum	

IP header

Version	Header length	Type of service	Total length
Identification	Flags	Fragment offset	
Time to live	Protocol	Header checksum	
Source address	Destination address		
Options			
Padding			

Version - v4/v6
 Protocol - icmp / tcp / udp
 TTL
 S/P
 D/P
 Fragment
 TOS = Type of service (QoS)

Data-link } 2 layer

RIP v1, v2
fism

Router rip
version 2 } CMD

V1 - 255.255.255.255
V2 - 224.0.0.9

